

Technical Data

Product Description

Unreinforced, Flame Retardant, Non-Halogenated, Elemental phosphorous free, Polyamide 66

General

Material Status	• Commercial: Active
Literature ¹	• Processing - Injection Molding (English) • Typical Processing for DuPont Engineering Polymers (English) • Typical Processing for DuPont Engineering Polymers (English)
UL Yellow Card ²	• E41938-234498
Search for UL Yellow Card	• DuPont Performance Polymers • Zytel®
Availability	• Africa & Middle East • Asia Pacific • Europe • Latin America • North America
Additive	• Flame Retardant • Mold Release
Features	• Flame Retardant • Halogen Free • Low (to None) Phosphorus Content
RoHS Compliance	• Contact Manufacturer
Forms	• Pellets
Processing Method	• Injection Molding
Part Marking Code (ISO 11469)	• >PA66-FR(30)<
Resin ID (ISO 1043)	• PA66-FR(30)
ISO Designation	• ISO 16396-PA66,FR(30),M1F1G1NR,S14-040

Physical	Nominal Value Unit	Test Method
Density	1.16 g/cm ³	ISO 1183
Molding Shrinkage		ISO 294-4
Across Flow	1.0 %	
Flow	0.90 %	
Water Absorption		ISO 62
24 hr, 23°C, 2.00 mm	1.8 %	
Equilibrium, 23°C, 2.00 mm, 50% RH ⁴	2.4 %	

Mechanical	Nominal Value Unit	Test Method
Tensile Modulus	3750 MPa	ISO 527-2
Tensile Stress (Break)	80.0 MPa	ISO 527-2
Tensile Strain (Break)	10 %	ISO 527-2
Flexural Modulus	3660 MPa	ISO 178
Poisson's Ratio	0.36	

Impact	Nominal Value Unit	Test Method
Charpy Notched Impact Strength		ISO 179/1eA
-40°C	3.0 kJ/m ²	
-30°C	3.0 kJ/m ²	
23°C	3.7 kJ/m ²	
Charpy Unnotched Impact Strength (23°C)	140 kJ/m ²	ISO 179/1eU

Thermal	Nominal Value Unit	Test Method
Heat Deflection Temperature		
0.45 MPa, Unannealed	235 °C	ISO 75-2/B
1.8 MPa, Unannealed	84.0 °C	ISO 75-2/A
Ball Pressure Test (220°C)	Pass	IEC 60695-10-2
Melting Temperature ⁵	260 °C	ISO 11357-3

Electrical	Nominal Value Unit	Test Method
Electric Strength	31 kV/mm	IEC 60243-1
Comparative Tracking Index	600 V	IEC 60112



Flammability	Nominal Value Unit	Test Method
Flame Rating		UL 94
0.75 mm	V-0	IEC 60695-11-10, -20
1.5 mm	V-0	
Oxygen Index	39 %	ISO 4589-2
FMVSS Flammability ⁶	DNI	FMVSS 302
Fill Analysis	Nominal Value Unit	
Ejection Temperature	210 °C	
Injection	Nominal Value Unit	
Drying Temperature	80 °C	
Drying Time - Desiccant Dryer	2.0 to 4.0 hr	
Suggested Max Moisture	0.20 %	
Processing (Melt) Temp	270 to 290 °C	
Melt Temperature, Optimum	280 °C	
Mold Temperature	50 to 90 °C	
Mold Temperature, Optimum	70 °C	
Holding Pressure	50.0 to 100 MPa	
Drying Recommended	yes	
Hold Pressure Time	3.00 s/mm	
Maximum Screw Tangential Speed	12 m/min	

Notes

¹ These links provide you with access to supplier literature. We work hard to keep them up to date; however you may find the most current literature from the supplier.

² A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

³ Typical properties: these are not to be construed as specifications.

⁴ Assessed

⁵ 10°C/min

⁶ Derived from Similar Grade



Where to Buy

Supplier**DuPont Performance Polymers**

Wilmington, DE USA

Telephone: 302-999-4592**Web:** <http://plastics.dupont.com/>

Distributor**Biesterfeld***Biesterfeld is a Pan European distribution company. Contact Biesterfeld for availability of individual products by country.***Telephone:** +49-40-32008-0**Web:** <http://www.biesterfeld-plastic.com/>**Availability:** Algeria, Austria, Belgium, Bosnia and Herzegovina, Brazil, Bulgaria, Croatia, Cyprus, Czech Republic, Egypt, France, Germany, Greece, Hungary, Italy, Libyan Arab Jamahiriya, Luxembourg, Mauritania, Morocco, Netherlands, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Switzerland, Tunisia, Turkey**CCC Plastics****Telephone:** 800-465-6917**Web:** <https://www.ccc-group.com/>**Availability:** Canada**Distrupol Ltd***Distrupol Ltd is a Pan European distribution company. Contact Distrupol Ltd for availability of individual products by country.***Telephone:** 08452003040**Web:** <http://www.distrupol.com/>**Availability:** Denmark, Finland, Ireland, Norway, Sweden, United Kingdom**PolyOne Distribution***PolyOne Distribution is a global distribution company. Contact PolyOne Distribution for availability of individual products by country.***Telephone:** 800-894-4266**Web:** <http://polyonedistribution.com/>**Availability:** Global